



Review

Human genetics and its impact on cardiovascular disease

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ABSTRACT

Cardiovascular disease (CVD) is a major cause of death worldwide. Given that CVD is a highly heritable trait, researchers have attempted to fully understand the genetic basis of CVD for a long time. The human genome comprises 3,100 Mbp per haploid genome and 6,200 Mbp in total (diploid genome). However, there is a tendency for rare genetic variations to exhibit a large effect size, whereas common genetic variations have a small effect on diseases, because of natural selection. In this sense, dividing genetic variations into two groups based on allele frequency (and effect sizes on diseases) is a good idea. We know there are several important genes (especially lipid-related genes) in which rare genetic variations are apparently associated with CVD risk, while a polygenic risk score comprising common genetic variations appears to work quite well among general populations. That information can be used not only for risk stratification but also for discoveries for novel pharmacologic targets. In this review article, we provide the important and simple idea that human genetics is important for CVD because it is a highly heritable trait, and we believe that it will lead to precision medicine in this field.

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Introduction

Cardiovascular disease (CVD) is one of the major causes of death worldwide [1]. A number of studies have been conducted to find ways to combat this disease. Among them, epidemiological studies have contributed greatly to our better understanding of backgrounds of CVD. For example, the Framingham Heart Study in the USA [2–4] as well as the Hisayama study in Japan [5,6] revealed many risk factors associated with CVD, such as cholesterol, hypertension, and smoking. Although epidemiological studies have revealed many factors associated with CVD, the causalities between those factors and CVD have been different stories. In this regard, randomized controlled trials have been conducted to show the efficacy of the intervention of factors identified through epidemiological studies [7,8]. On the other hand, it is well known that CVD is a highly heritable trait; thus, it is reasonable to account for genetic factors in the clinical practice of CVD [9]. Family history information has long been considered as a way to achieve this end, and studies have shown that family history information operates quite well for CVD risk assessment [10–12]. However, “family history” is a type of binary trait that is difficult to assess quantitatively and qualitatively. Recent advances in genetics have enabled us to quickly see the human DNA sequence and have illuminated

genes associated with CVD development [13]. By investigating the associations between the human genome and CVD in detail, we can identify high-risk individuals in the early phase in life, determine precision medicine methods based on their causes, and discover novel therapeutic approaches. In this review article, we provide important information on the impact of human genetics for CVD and how to translate this information into our daily clinical practice of CVD.

Human genome

The human genome comprises 3,100 Mbp per haploid genome and 6,200 Mbp in total (diploid genome). Although the sequence of the human genome has been nearly fully determined by DNA sequencing, it is not yet fully understood. Moreover, much more work needs to be done to further elucidate the biological functions of their protein and RNA products. More importantly, recent studies suggest that most of the vast quantities of noncoding DNA within the genome have associated biochemical activities, including regulation of gene expression, organization of chromosome architecture, and signals controlling epigenetic inheritance [14–16]. It is impossible to understand the human genome as it is. Therefore, a useful scheme has been proposed to understand the relationship between the human genome and disease, namely, to divide rare and common genetic variations [17]. Rare genetic variations tend to have larger effects on disease as compared with common genetic variations. This hypothesis has been well documented

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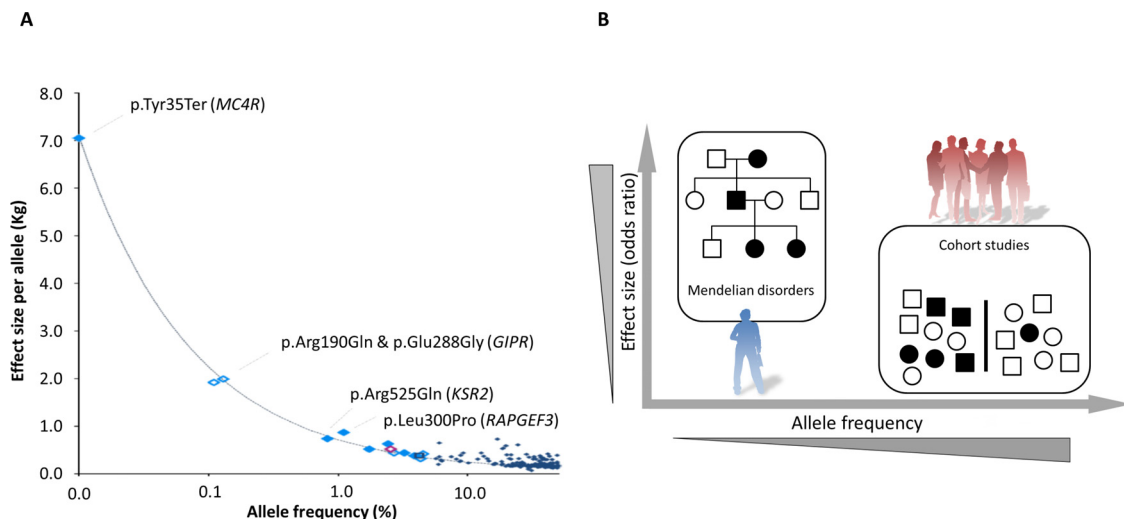


Fig. 1. Relationship between allele frequency and effect size. (A) Findings from an association study on body weight. (B) Relationships between allele frequency and effect size. *MC4R*, melanocortin 4 receptor; *GIPR*, gastric inhibitory polypeptide receptor; *KSR2*, kinase suppressor of Ras 2; *RAPGEF3*, Rap guanine nucleotide exchange factor.

in the case of body weight [18] (Fig. 1). It is not hard to imagine that body weight is also a highly heritable trait. We performed an association study investigating genetic backgrounds associated with body weight and identified an extremely rare genetic variation whose effect on body weight is as much as 7 kg per allele, whereas there are many common genetic variations whose effects on body weight are about ~100 g per allele. From a clinical point of view, a rare genetic variation with an effect of 7 kg is important, whereas common genetic variations with a ~100 g effect are negligible. However, if we can concatenate the effects via multiple common genetic variations, then such a “score” may be important enough for our daily clinical practice.

CVD is a heritable trait

How about CVD? Is it a heritable trait with genetic information attractive for researchers and clinicians? The answer is yes. Epidemiological studies have shown that information about family history is significantly associated with CVD, suggesting that CVD is a heritable trait. Based on data from a twins study, the degree of heritability of CVD appears to be as high as ~50% [19]. In accordance with these data, a current clinical guideline on preventive atherosclerotic CVD (ASCVD) in Japan strongly suggests assessing family history information [20].

Rare genetic diseases associated with an increased risk for CVD via serum lipids

Among several rare genetic diseases associated with increased CVD risk, inherited lipid disorders are the most important to note (Fig. 2). Familial hypercholesterolemia (FH), caused by genetic mutation(s) in the low-density lipoprotein (LDL) receptor or its associated gene, is one of the most important diseases in this context [21–26]. Patients with FH typically exhibit extreme elevation of LDL cholesterol (LDL-C), tendon xanthomas, and premature CVD. Although this disease is categorized as a “rare” disease, the prevalence among the general population has been estimated to be 1 in 300 [27,28]. More importantly, patients with FH have extremely high risk for CVD if untreated, although early intervention via statins can offset their risk [29,30]. Thus, we must pay much more attention to the detection of FH. In addition to FH, we found that individuals who have a rare (1 in 1,000 individuals) damaging mutation in ATP-binding cassette subfamily G member 5

(*ABCG5*) exhibited increased risk for CVD [31]. On the other hand, triglyceride-rich lipoproteins have been implicated as a causal factor for CVD [32,33]. For example, 1 in 250 individuals with elevated triglycerides via a damaging mutation in lipoprotein lipase (*LPL*) has a 1.8-fold increased odds for CVD [34]. In addition, 1 in 3,000 individuals with elevated triglycerides via a damaging mutation in apolipoprotein A5 (*APOA5*) has 4.5-fold increased odds for CVD [35]. Finally, individuals with a particular single-nucleotide polymorphism in lipoprotein (a) (*LPA*) (1 in 50 individuals) exhibited increased serum Lp(a) and increased odds for CVD (odds ratio = 1.92). These are obvious examples of inherited rare genetic disorders associated with CVD [36].

Cardioprotective “human knockout”

In addition to “disorders” associated with an increased risk for CVD, there are several inherited situations in which rare genetic variations contribute to protection from CVD. For example, 1 in 150 individuals with reduced triglyceride levels via a damaging mutation in apolipoprotein C3 (*APOC3*) has exhibited reduced risk for CVD (odds ratio = 0.60) [37]. Importantly, individuals with *APOC3* knocked out displayed marked blunting of the postprandial increase of triglycerides [38]. Similarly, 1 in 300 individuals who exhibited combined hypolipidemia via a damaging mutation in angiopoietin-like 3 (*ANGPTL3*) has exhibited a reduced risk for CVD (odds ratio = 0.61) [39]. In addition, we have shown that individuals with a damaging mutation in cholesteryl ester transfer protein (*CETP*) exhibited a reduced CVD risk, probably because of their reduced LDL-C levels rather than increased high-density lipoprotein cholesterol levels [40]. Other examples of cardioprotective human knock out include familial hypobetalipoproteinemia and deficiency in microsomal triglyceride transfer protein (MTTP). We previously showed that individuals with familial hypobetalipoproteinemia with loss-of-function mutation in apolipoprotein B (*APOB*) exhibited extremely lower CVD risk, probably because of reduced atherogenic *APOB*-containing lipoproteins, including LDL-C, triglycerides, and Lp(a) altogether [41]. Moreover, we have shown that patients with MTTP deficiency exhibited a protective effect for a postprandial increase in remnant lipoprotein, in addition to extremely low LDL-C levels [42]. Finally, in our daily clinical practice, we are now using ezetimibe [an inhibitor for Niemann–Pick C1-Like 1 (NPC1L1)] and a proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor in accordance with the findings from

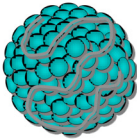
Gene	Frequency	Biomarker		Increased odds ratio for CVD
- LDL receptor - PCSK9 - Apolipoprotein B	1/300		LDL-C	3.2-fold
ABCG5	1/1,000		LDL-C	2.0-fold
- LPL - Apolipoprotein A5	1/500 1/3,000		TG-rich lipoprotein	1.8-fold 4.5-fold
Lipoprotein (a)	1/100		Lp(a)	2.8-fold

Fig. 2. Rare genetic diseases associated with increased risk for CVD via serum lipids. Genes, frequencies, biomarkers, and their impacts on CVD are shown. CVD, cardiovascular disease; PCSK9, proprotein convertase subtilisin/kexin type 9; ABCG5, ATP-binding cassette sub-family G member 5; LPL, lipoprotein lipase; LDL-C, low-density lipoprotein cholesterol; TG, triglycerides.

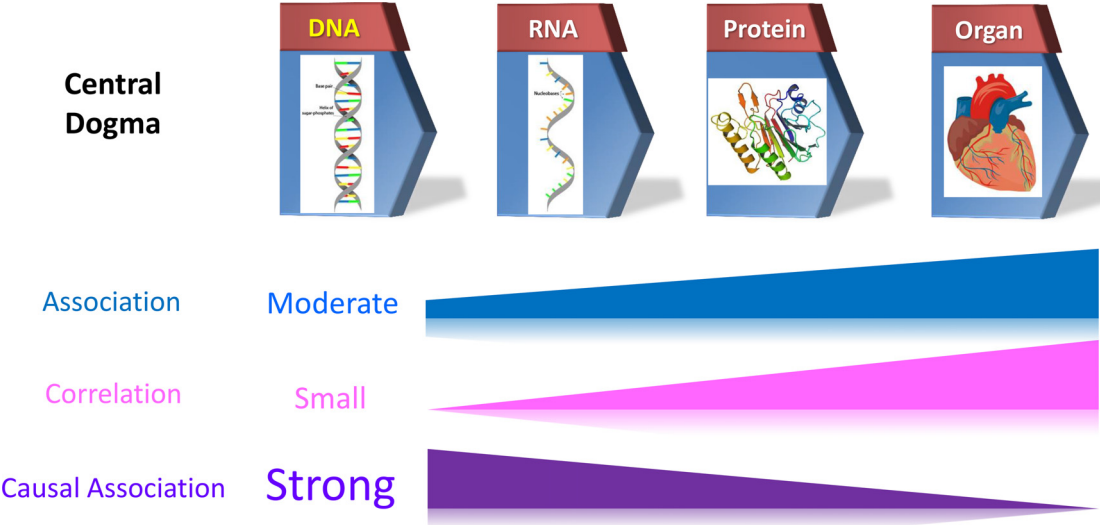


Fig. 3. Association, correlation, and causal association with disease. The upper panel indicates the central dogma of molecular biology. The lower panels show the degree of association, correlation, and causal association with disease.

the human knockouts [43–45]. It is important to note that there is a strong causal association between DNA variant and outcome (Fig. 3). These findings have led to the development of novel pharmacologic therapies targeting a reduction in CVD (Table 1).

Common genetic variations and CVD: the idea of a polygenic risk score

In contrast to rare genetic variations, the effects of common genetic variations on disease are typically small. Given that genome-wide association studies (GWAS) have revealed many common genetic variations associated with CVD [46–48], researchers have come up with the idea that an aggregation of common genetic variations into a single score might be clinically useful. The initial attempt at developing a polygenic risk score (PRS) was published in 2010. Those researchers used a PRS based on 13 single-nucleotide variations (SNVs) associated with GWAS of European ancestries (case-control design and prospective cohort) and then

tested whether the PRS was associated with ASCVD events among independent cohorts. They found a significant association between the PRS and ASCVD events; however, at that time, they failed to show its usefulness in receiver operating characteristic (ROC) and net reclassification improvement analysis [49]. In 2015, Mega et al. increased the number of common genetic variations to 27 to determine whether a PRS outperforms reclassification using traditional risk factors. They found that a PRS comprising 27 common genetic variations surpassed its threshold [50]. In addition, we compared the power of a PRS comprising 27 SNVs and that of a PRS containing 50 SNVs and found that the larger number of SNVs increases its predictive power, independent of family history information [51]. Simultaneously, we showed that a PRS comprising 12 SNVs associated with atrial fibrillation was significantly associated with ischemic stroke events (probably cardioembolic stroke) [52]. Furthermore, Khera et al. confirmed that a PRS using approximately 6 million SNVs to predict ASCVD was significantly associated with actual ASCVD events in a score-dependent manner [53].

Table 1
Novel pharmacologic interventions to lower LDL levels.

Target	Deficiency or carriers of PTV	Compounds	Affected lipids	Mendelian randomization
ANGPTL3	Familial combined hypolipoproteinemia	Evinacumab	LDL-C ↓↓ Triglycerides ↓↓	Ref 42
APOB	Familial hypobetalipoproteinemia	Mipomersen	LDL-C ↓↓ Triglycerides ↓↓ Lp (a) ↓↓	Ref 44
APOC3	APOC3 deficiency	Volanesorsen	Triglycerides ↓↓↓	Ref 37
CETP	CETP deficiency	Anacetrapib	HDL-C ↑↑↑ LDL-C ↓	Ref 43
MTTP	Abetalipoproteinemia	Lomitapide	LDL-C ↓↓↓	NA
NPCL1	Heterozygous carriers (1 in 650 individuals)	Ezetimibe	LDL-C ↓	Ref 38
PCSK9	Familial hypobetalipoproteinemia	Evolocumab Alirocumab	LDL-C ↓↓↓ Lp(a) ↓	Ref 39

ANGPTL3, angiopoietin-like 3; APOB, apolipoprotein B; APOC3, apolipoprotein C3; CETP, cholesteryl ester transfer protein; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; MTTP, microsomal triglyceride transfer protein; NPC1L1, Niemann–Pick C1-Like 1; PCSK9, proprotein convertase subtilisin-kexin type 9; NA, not available; Ref, reference.

Table 2
Studies of PRS for the prediction of CVD.

Authors [reference]	Year of publication	PRS	Population	Outcomes	Main results
Ripatti S et al. [49]	2010	13 SNV	Finnish and Swedish cohorts; (Caucasian)	Coronary heart disease	HR 1.66 (95% CI 1.35–2.04) Top vs. bottom quintile (PRS)
Tada H et al. [52]	2014	12 SNV	Swedish cohort; (Caucasian)	Ischemic stroke	HR 1.23 (95% CI 1.04–1.46) Top vs. bottom quintile (PRS)
Mega JL et al. [50]	2015	27 SNV	Swedish cohort; (Caucasian) Primary/Secondary prevention groups in US; (most Caucasian)	Coronary heart disease	HR 1.81 (95% CI 1.22–1.47) Top vs. bottom quintile (PRS)
Tada H et al. [51]	2016	27 & 50 SNV	Swedish cohort; (Caucasian)	Coronary heart disease	HR 1.70 (95% CI 1.48–1.94) for 27-SNV-PRS HR 1.92 (95% CI 1.67–2.20) for 50-SNV-PRS Top vs. bottom quintile (PRS)
Khera AV et al. [53]	2018	6.6 M SNV	UK Biobank; (Caucasian)	Coronary heart disease	OR 4.53 (95% CI 3.95–5.17) Top 1% vs. remaining 99%
Hachiya T et al. [54]	2020	>350,000 SNV	Japanese cohort; (East Asian)	Ischemic stroke	HR 2.44 (95% CI 1.16–5.12) Top vs. bottom quintile (PRS)
Wang M et al. [55]	2020	6.6 M SNV	UK Biobank; (South East Asian) Case-control study in Bangladesh and India; (South East Asian)	Coronary heart disease	OR 2.46~4.16 Top 5% vs. middle quintile (PRS)
Koyama S et al. [56]	2020	~1M SNV	Biobank Japan; (East Asian) UK Biobank; (Caucasian) C4D consortium; (Caucasian)	Coronary heart disease	OR 2.65 (95% CI 2.30–3.05) Top 10% vs. remaining 90%

CVD, cardiovascular disease; PRS, polygenic risk score; SNV, single-nucleotide variations; US, United States; UK, United Kingdom; HR, hazard ratio; CI, confidence interval; OR, odds ratio.

Table 2 summarizes the attempts to predict ASCVD events using a PRS. Thus far, a couple of studies have been conducted to determine whether a PRS is similarly useful among Asian populations. Hachiya et al. reported that a genome-wide PRS comprising >350,000 common generic variations was associated with ischemic stroke events among the general Japanese population, independent of other environmental factors [54]. They used GWAS summary statistics from the BioBank Japan Project, Tohoku Medical Megabank Project, Japan Public Health Center-based prospective study, and Japan Multi-Institutional Collaborative Cohort Study and assessed the ischemic stroke events in the Hisayama study, which is a prospective population-based study that was initiated in 1961 to investigate the epidemiology of ASCVD in a small town on Kyushu Island, Japan. Wang et al. showed that a genome-wide PRS comprising >6 million common genetic variations was associated with coronary artery disease (CAD) among South Asian (Indian and Bangladeshi) populations [55]. Interestingly, they essentially used GWAS summary statistics from European ancestry to make a genome-wide PRS and adjusted it for ancestry using a set of whole-genome sequencing data from South Asian individuals. These results indicate that the cumulative impact of common DNA variants, which is now possible to quantify using a PRS, is an important driver of CAD risk, even among individuals of South Asian ancestry. By optimizing a polygenic score for CAD in the South Asian population, we noted a 3.22- to 3.91-fold increase in risk when comparing the highest to lowest quintiles across three independent study samples. Moreover, the pattern of disease associ-

ations was strikingly concordant across individuals of South Asian ancestry living in the United Kingdom, Bangladesh, and India, with an odds ratio/standard deviation increment ranging from 1.58 to 1.66 across the three studies. These results suggest the feasibility of the transfer of polygenic scores across varying environmental exposures.

Most recently, Koyama et al. showed that a transethnic genome-wide PRS outperformed an ethnicity-specific risk score in predicting CAD among Japanese populations [56]. They performed a transethnic meta-analysis combining Japanese GWAS (mainly using BioBank Japan), CardiogramC4D (C4D), and UK Biobank and made a transethnic genome-wide PRS. Moreover, the most common genetic variations with a nominal significance showed a consistent direction of effect between Japanese and European ancestries.

Clinical usefulness of PRS

In addition to risk stratification beyond traditional clinical risk factors, PRS has been shown to be a useful tool for various purposes. First, among individuals with high PRS, the effectiveness of statins for CVD appears to be far greater than among those with low PRS for CVD [57]. This is great news, because it has been shown that the effectiveness of statins is equal among all classic subgroups, such as diabetes, hypertension, and gender [58]. Second, among individuals with a high PRS for CVD, a healthy lifestyle is associated with a reduced risk for CVD, suggesting that such inherited CVD risk appears to be offset by healthy lifestyle [59].

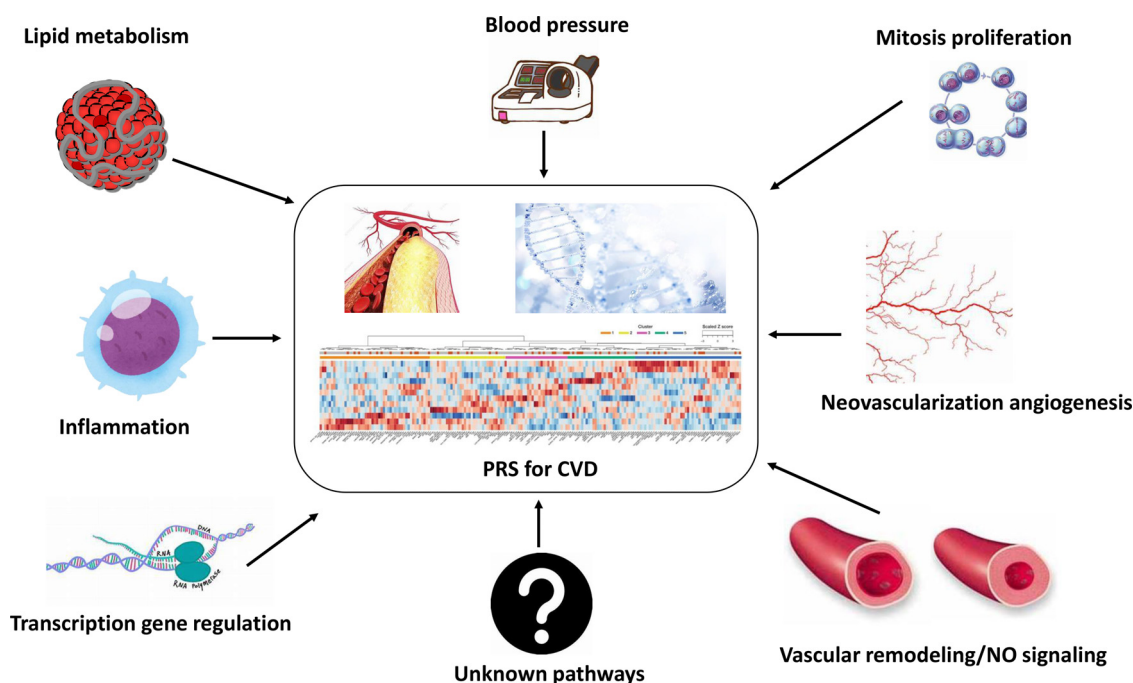


Fig. 4. Pathways comprising PRS for CVD. Several different pathways are contributing to increase or decrease life-long risk for CVD. PRS, polygenic risk score; CVD, cardiovascular disease; NO, nitric oxide.

Third, such PRSs comprising common SNVs were significantly associated with ASCVD among patients with FH in whom a rare genetic variation in the LDL receptor or its associated gene elevated the risk of ASCVD [60].

Limitations and future perspectives of PRS

A PRS is useful for risk prediction because it is a normally distributed continuous number, regardless of causal pathways. Alternatively, however, which elements contribute to the elevation of the PRS should be important. For example, when we encounter two individuals whose PRSs appear to be similarly high, but the cause of the elements is different (e.g. individual A has many risk alleles in inflammatory-associated variants, and individual B has many risk alleles in lipid metabolism), then antiinflammatory agents may be more effective in individual A, whereas lipid-modification drugs may be better for individual B (Fig. 4). Moreover, as has been shown previously, a particular behavior may be associated with the offset of elevated PRS. Lifestyle and dietary habits are different across regions and ethnicities; thus, much more data are needed to elucidate “good” behavior among particular populations according to a PRS and its elements.

In addition, most GWAS conducted thus far have used logistic models that cannot capture the interacting associations between genetic variations and an outcome, which might account for the so-called “missing heritability.” In this sense, more sophisticated statistical models are needed to identify such associations, which in turn can be used as instruments to develop a more powerful PRS.

How to translate this information to our daily clinic?

We need to keep in mind that there are Mendelian disorders associated with CVD. In that case, cascade screening is a useful way to diagnose affected relatives early in their life. In the case of FH, we have shown that cascade screening for FH was significantly associated with the affected relatives’ prognosis. Moreover, there is an example in which the existence of a particular type of

mutation affects the effect by a pharmacologic intervention [61]. Moreover, we would like to emphasize that at least some patients exhibited an extremely early myocardial infarction associated with sitosterolemia [62–66], which is now considered to be much more frequent among the general population. Second, inherited CVD risk via common genetic variations (high PRS for CVD) appears to be concordant within their relatives [67]. This suggests that “cascade screening” for high-risk individuals should include individuals with high PRS for CVD. It is of note that PRS for CVD is associated with CVD independent of family history information.

How about other phenotypes in cardiology?

We focused on ASCVD in this review article. However, the principles we demonstrated here are applicable to other phenotypes in cardiology, such as arrhythmias, cardiomyopathies, and aortic disease, as they are also highly heritable traits [68–72].

Conclusion

We now know CVD is a highly heritable trait. Both rare and common genetic variations are useful not only for risk stratification but also for cascade screening. Genetic information is not a stigma, because we can offset its effects by early diagnosis and appropriate interventions. We need to organize randomized controlled trials to determine the actual usefulness of such genetic information to implement this important element into our clinical practice.

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The authors declare that there is no conflict of interest.

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